

1. Project Overview

This project analyses employee data to identify patterns associated with employee attrition and understand the factors that may influence workforce retention.

The analysis examines employee demographics, job roles, compensation, overtime, job satisfaction, work-life balance, environment satisfaction, promotion history, and other workforce characteristics.

The objective is to transform employee data into practical HR insights that can help organizations identify higher-risk employee groups and develop targeted retention strategies.

2. Business Problem

Employee turnover can increase recruitment costs, create workforce gaps, and affect team productivity. HR teams therefore need to understand which employee groups have higher attrition rates and which workplace factors are associated with employees leaving the organization.

This analysis focuses on questions such as:

- How high is the overall employee attrition rate?
- Is overtime associated with higher attrition?
- Which job roles and departments have higher attrition?
- How are job satisfaction and work-life balance related to attrition?
- Does income level show differences in employee attrition?
- Which employee groups may require greater retention attention?

The goal is not only to measure attrition, but to identify actionable workforce patterns that can support HR retention decisions.

3. Business Questions

The analysis aims to answer the following business questions:

1. What is the overall employee attrition rate, and which employee groups have higher attrition?
2. Is overtime associated with a higher employee attrition rate?
3. How does job satisfaction relate to employee attrition?
4. Does work-life balance show differences in employee attrition?
5. Is employee income level associated with different attrition rates?
6. Which departments and job roles have the highest attrition rates?
7. Which workforce factors and employee groups should HR prioritize for retention efforts?

4. Dataset

The project uses the IBM HR Analytics Employee Attrition & Performance dataset, containing employee-level information related to demographics, job roles, compensation, satisfaction, overtime, performance, and attrition.

- Employees: 1,470
- Columns: 35
- Main outcome: Employee Attrition
- Data type: Employee-level HR records
- Source: IBM HR Analytics Employee Attrition & Performance dataset

5. Tools & Technologies

Microsoft Excel — Data cleaning, validation, PivotTables, PivotCharts, calculations, XLOOKUP, and dashboard development.

Excel Dashboard — KPI tracking and interactive analysis using PivotCharts and slicers.

6. Data Preparation

The IBM HR Employee Attrition dataset was prepared for employee attrition analysis using Excel-based data validation, transformation, and analytical grouping.

The main preparation steps included:

- Reviewed the dataset structure, employee records, and available variables.
- Checked for duplicate employee records using a Duplicate_flag validation field.
- Checked for blank or missing values using blank_flag_all.
- Validated categorical fields such as Department, Job Role, Business Travel, Marital Status, Gender, and Attrition.
- Created standardized/lookup fields to make coded variables easier to interpret, including:
 - Education
 - Environment Satisfaction
 - Job Involvement
 - Job Level
 - Job Satisfaction
 - Performance Rating
 - Relationship Satisfaction
 - Stock Option Level
 - Work-Life Balance
- Created analytical groups for variables such as:
 - Age
 - Monthly Income
 - Distance From Home
 - Years at Company
 - Total Working Years
- Created employee career-status fields, including promotion status.
- Created helper fields for attrition analysis, including Attrition_Yes and overtime-related analysis fields.
- Validated numeric variables for inappropriate or inconsistent values.
- Prepared the dataset for Pivot Table analysis and dashboard visualization.
- Focused the analysis on identifying patterns associated with employee attrition rather than assuming direct causation.

Data Preparation Workflow

Raw IBM HR Dataset

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Data Quality Checks

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Duplicate & Blank Validation

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Categorical Value Validation

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Standardization & Lookup Fields

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Analytical Group Creation

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Helper Columns

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Pivot Table Analysis

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Dashboard Development

7. Analysis Approach

The analysis was designed to answer the following business question:

Why are employees leaving the organization, and which employee, job, compensation, and workplace factors are associated with higher attrition?

Rather than analyzing every column independently, the analysis was structured around major HR business areas and compared attrition rates across employee groups.

Analysis Workflow

Validated HR Dataset

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Employee Attrition Analysis

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Employee Segmentation

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Department & Job Role Analysis

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Workplace & Satisfaction Analysis

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Compensation & Career Analysis

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Attrition Rate Comparison

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Key Findings

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Business Interpretation

7.1 Overall Attrition

The analysis first established the overall employee attrition level using:

- Total Employees
- Total Attrition
- Attrition Rate
- Average Years at Company

This provides the baseline against which individual employee groups can be compared.

7.2 Department and Job Role

Attrition was compared across:

- Department
- Job Role

This helps identify departments and specific roles where employee retention may require further investigation.

7.3 Workplace Factors

The analysis examined:

- Overtime
- Work-Life Balance
- Job Satisfaction
- Environment Satisfaction
- Business Travel

These variables were used to identify workplace conditions associated with higher attrition.

7.4 Compensation and Career Factors

The analysis also compared attrition across:

- Monthly Income
- Promotion Status
- Job Level
- Years at Company
- Years in Current Role
- Years Since Last Promotion
- Years With Current Manager

These factors help investigate whether compensation and career progression may be associated with employee turnover.

7.5 Employee Characteristics

Additional segmentation was performed using:

- Age Group
- Gender
- Marital Status
- Distance From Home

These variables provide demographic and employee-context perspectives on attrition.

Analytical Principle

The analysis identifies relationships and patterns in the dataset. These relationships should not automatically be interpreted as proof that a particular factor directly caused an employee to leave.

8. Dashboard

8.1 HR Employee Attrition Analysis

Business Question

Why are employees leaving the organization, and which employee, job, workplace, compensation, and career factors are associated with higher attrition?

Dashboard Overview

The IBM HR Employee Attrition Dashboard provides an overview of employee turnover and highlights factors associated with higher attrition.

The dashboard is organized into three main analytical areas:

Overall Attrition

Provides the overall HR situation through:

- Total Employees
- Total Attrition
- Average Years at Company
- Attrition Rate
- Attrition by Department
- Attrition by Job Role

Possible Reasons

Examines potential factors associated with higher attrition:

- Overtime
- Work-Life Balance
- Job Satisfaction
- Monthly Income
- Environment Satisfaction
- Promotion Status

Employee Segmentation

Interactive filters allow attrition patterns to be explored by:

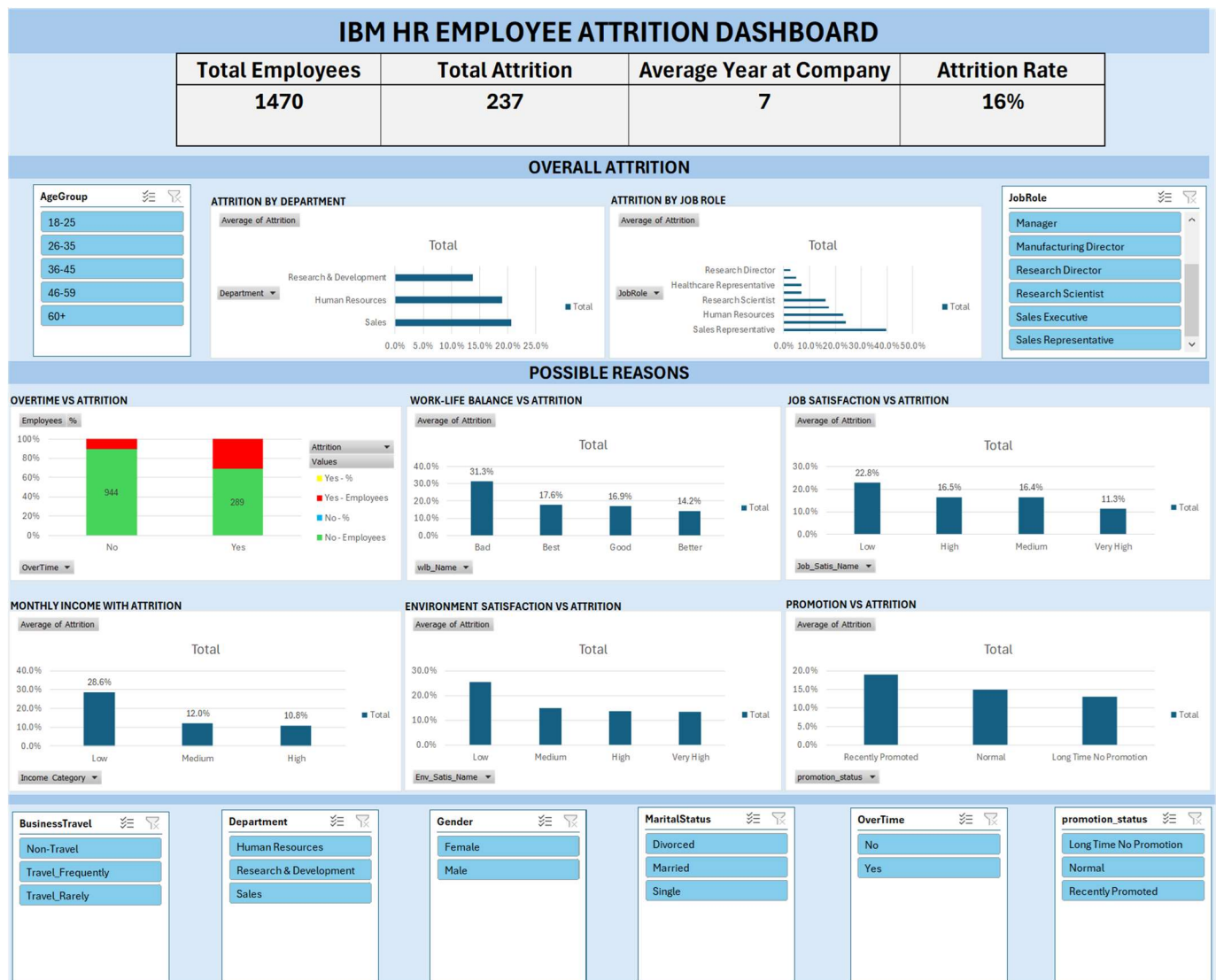
- Age Group
- Department
- Job Role
- Business Travel

- Gender
- Marital Status
- Overtime
- Promotion Status

This allows HR stakeholders to move from the overall attrition picture to specific employee segments.

8.2 Dashboard Screenshot

IBM HR Employee Attrition Dashboard



8.3 Key Findings

The dashboard highlights several notable attrition patterns, including higher attrition among employees working overtime, employees with low job satisfaction, employees reporting poor work-life balance, lower-income employees, and selected departments and job roles.

9. Key Insights

9.1 Overall Attrition

The dataset contains **1,470 employees**, of whom **237 employees left the organization**, resulting in an overall attrition rate of **16.12%**.

The average employee tenure at the company is approximately **7.01 years**.

This establishes the overall workforce retention baseline for the analysis.

9.2 Overtime & Attrition

Overtime shows one of the clearest differences in attrition.

- **Overtime employees:** 30.53% attrition
- **Non-overtime employees:** 10.44% attrition
- **Overall:** 16.12%

Employees working overtime experienced an attrition rate almost **three times** that of employees who did not work overtime.

Business insight: Overtime is strongly associated with higher employee attrition and may indicate workload or work-pressure areas that HR should investigate.

9.3 Job Satisfaction & Attrition

Attrition varied across job satisfaction levels:

- **Low:** 22.84%
- **High:** 16.52%
- **Medium:** 16.43%
- **Very High:** 11.33%

Employees reporting low job satisfaction had the highest attrition rate, while employees with very high job satisfaction had the lowest.

Business insight: Lower job satisfaction is associated with increased employee attrition, suggesting that employee experience should be monitored as part of retention planning.

9.4 Work-Life Balance & Attrition

Work-life balance also showed meaningful differences:

- **Bad:** 31.25%
- **Best:** 17.65%
- **Good:** 16.86%
- **Better:** 14.22%

Employees in the **Bad** work-life balance group had the highest attrition rate.

Business insight: Poor work-life balance is associated with substantially higher attrition and may warrant further investigation into workload, scheduling, and employee well-being.

9.5 Monthly Income & Attrition

Attrition differed considerably by income category:

- **Low income:** 28.61%
- **Medium income:** 12.03%
- **High income:** 10.80%

The low-income group had a substantially higher attrition rate than the medium- and high-income groups.

Business insight: Compensation level is associated with differences in attrition. HR may consider reviewing compensation competitiveness, particularly among lower-income employee groups.

9.6 Environment Satisfaction & Attrition

Environment satisfaction showed the following attrition rates:

- **Low:** 25.35%
- **Medium:** 14.98%

- **High:** 13.69%
- **Very High:** 13.45%

Employees with low environment satisfaction had the highest attrition rate.

Business insight: A less satisfactory workplace environment is associated with higher employee attrition and should be considered alongside other employee experience factors.

9.7 Department & Job Role Attrition

Department

Attrition rates by department were:

- **Sales:** 20.63%
- **Human Resources:** 19.05%
- **Research & Development:** 13.84%

Sales had the highest department-level attrition.

Job Role

The highest job-role attrition rates were:

- **Sales Representative:** 39.76%
- **Laboratory Technician:** 23.94%
- **Human Resources:** 23.08%
- **Sales Executive:** 17.48%
- **Research Scientist:** 16.10%

Business insight: Attrition is not evenly distributed across the organization. Certain departments and roles, particularly Sales Representative, show substantially higher attrition and may require more targeted retention attention.

9.8 Promotion & Attrition

The analysis also compared attrition across the project's promotion-status categories:

- **Recently Promoted:** 18.93%
- **Normal:** 14.96%

- **Long Time No Promotion:** 13.08%

The "Recently Promoted" group shows the highest attrition rate in this analysis.

Important interpretation: The promotion-status categories are based on the **YearsSinceLastPromotion** field. A value of zero does not necessarily prove that an employee was recently promoted; it can also mean that no promotion has occurred since joining the company.

Therefore, this result should be treated as a **pattern for further investigation**, rather than evidence that promotion itself causes higher attrition.

10. Business Recommendations

Based on the identified patterns, the following actions could support employee retention:

1. Review Overtime & Workload

Investigate teams and roles with high overtime levels to understand workload pressure, staffing gaps, and working-hour patterns.

2. Improve Work-Life Balance

Review workload distribution, scheduling, and employee support programs for groups reporting poor work-life balance.

3. Monitor Employee Satisfaction

Use regular employee feedback and engagement surveys to identify departments or roles where job satisfaction is consistently low.

4. Review Compensation

Evaluate compensation competitiveness for lower-income employee groups, particularly where low income is combined with other higher-risk indicators.

5. Focus on High-Attrition Roles

Prioritize retention initiatives for high-attrition roles such as Sales Representatives and Laboratory Technicians.

6. Investigate Department-Level Differences

Sales and Human Resources show higher department-level attrition than Research & Development. HR should investigate whether workload, compensation, career opportunities, or employee experience contribute to these differences.

7. Use Multiple Factors for Retention Decisions

HR should avoid relying on a single variable when identifying employees or groups at higher attrition risk. Overtime, satisfaction, income, work-life balance, job role, and other employee characteristics should be considered together.

11. Conclusion

This analysis found an overall employee attrition rate of 16.12% across 1,470 employees.

The strongest differences in attrition were observed among employees working overtime, employees with low job satisfaction, employees with poor work-life balance, and employees in the low-income category.

Attrition also varied significantly across departments and job roles, with Sales Representatives showing the highest job-role attrition rate at 39.76%.

These findings suggest that employee retention is influenced by multiple workforce patterns rather than a single factor. The analysis provides HR teams with a data-driven starting point for identifying higher-risk groups and prioritizing areas for further investigation.

Because this is observational employee data, the findings represent associations rather than proven causes. Further HR investigation and additional employee-level information would be required to determine the underlying causes of attrition.

12. Skills Demonstrated

12.1 Excel Data Analysis

- Data cleaning
- Data validation
- Data transformation
- Helper-field creation
- XLOOKUP
- Conditional analysis
- KPI calculations

12.2 PivotTable & PivotChart Analysis

- Building PivotTables

- Calculating attrition rates
- Comparing employee groups
- Department and job-role analysis
- Factor-based workforce analysis
- Creating PivotCharts

12.3 Excel Dashboard Development

- KPI cards
- Interactive charts
- Slicers
- Dashboard layout
- Business-focused data visualization

12.4 HR & Business Analysis

- Employee attrition analysis
- Workforce segmentation
- Retention analysis
- Identifying high-risk employee groups
- Translating data into business insights

12.5 Data Quality & Validation

- Missing-value checks
- Duplicate checks
- Data consistency checks
- Categorical-value validation
- Numeric-value validation
- Lookup validation

12.6 Problem Solving

- Translating HR problems into analytical questions
- Selecting relevant variables

- Comparing employee groups
- Identifying meaningful patterns
- Evaluating limitations of the analysis

12.7 Data Storytelling & Communication

- Presenting analytical findings clearly
- Connecting metrics to business questions
- Communicating actionable insights
- Developing recommendations based on data
- Presenting results through an interactive Excel dashboard